11.H1.4 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about inscribing polygons in circles.

Unit 11, Lesson 3: Solving Problems Involving Triangles Inscribed in Circles

Benchmark

MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.

Learning Target

- ☐ I can solve mathematical problems involving triangles inscribed in a circle.

11.3.1 Warm-Up: Patterns

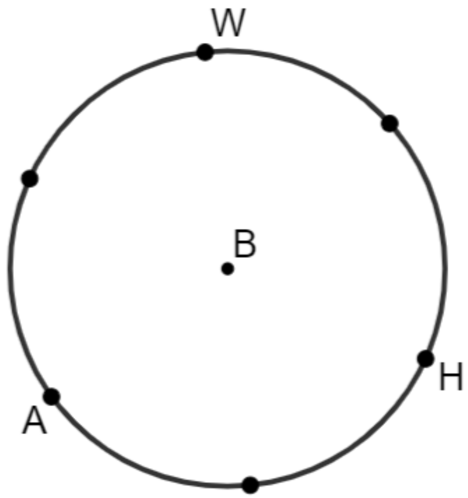
1. What is the first question that comes to your mind?

2. Describe the pattern used in the image.

11.3.2 Exploration: Triangles Inscribed in Circles

Work with your partner to complete the following.

Elijah is constructing a triangle inscribed in Circle B and stopped after he created six points on Circle B that are the same distance from each other.



1. Using a straightedge, draw a line from point A to point H.

2. Using a straightedge, draw a line from point H to point W.

3. In relation to the circle, what type of angle is $\angle AHW$?

4. Explain the relationship between $m\angle AHW$ and $m\widehat{WA}$.

5. Using a straightedge, draw a line from point A to point W.

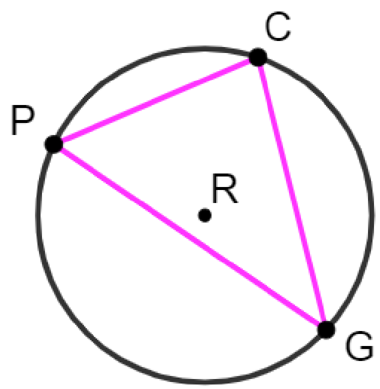
6. Use a ruler or patty paper to verify that ΔAHW is an equilateral triangle.

7. What is the $m\angle AHW$?

8. What is the $m\widehat{WA}$?

11.3.3 Guided Practice: Triangles Inscribed in Circles

1. Circle R is shown. Points P, C, and G are on circle R and create ΔPCG. $m\widehat{PG} = 162°$, $m\angle CPG = 58°$, and $m\angle PCG = (7x - 17)°$.

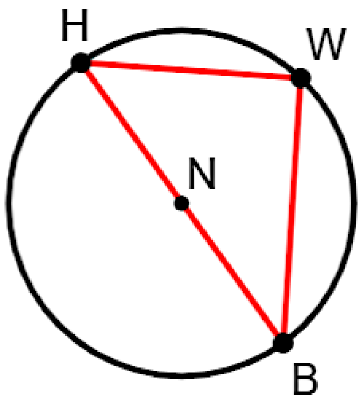


   a. What is the $m\angle PCG$?

   b. Find the value of x.

   c. What is the $m\angle PGC$?

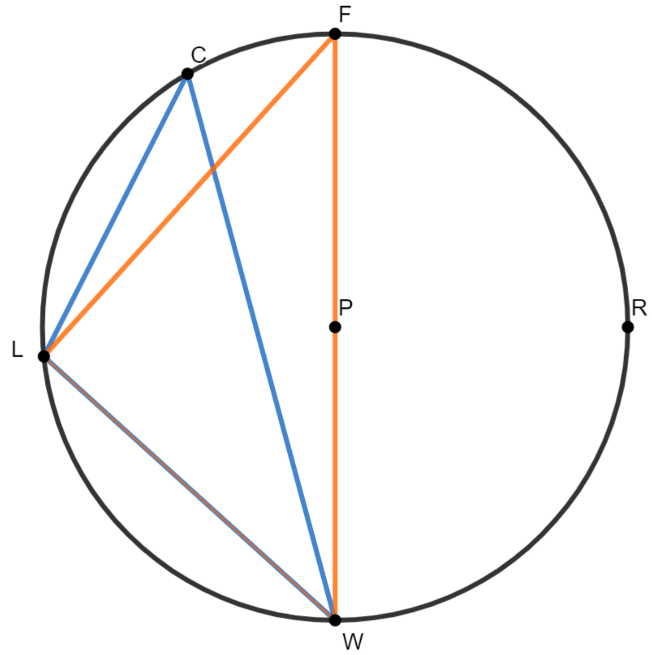
2. Circle N has points H, W, and B on the circle and diameter $\overline{HB}$. ΔHWB is inscribed in circle N. $m\,\widehat{BW} = 102°$ and $m\angle HBW = (4x - 10)°$.



a. What is the $m\angle HBW$?

b. Find the value of x.

3. ΔCLW and ΔFLW are inscribed in Circle P, where $\overline{FW}$ is the diameter of Circle P. $m\,\widehat{FC} = (2x - 6)°$, $m\,\widehat{LC} = (4x - 6)°$, and $m\,\widehat{LW} = (3x + 30)°$.

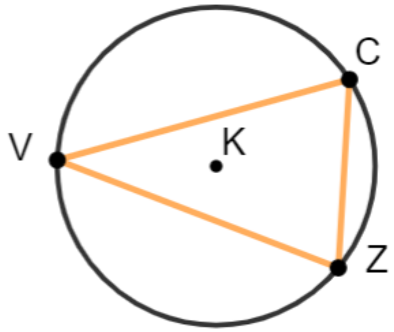


Complete the table.

Angle Measures	Angle Measures
$m\angle LCW =$	$m\angle LFW =$
$m\angle CWL =$	$m\angle FWC =$
$m\angle FLC =$	$m\angle FLW =$

11.3.4 Your Turn!

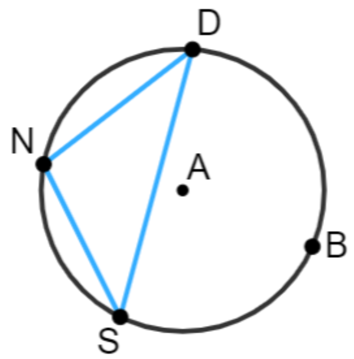
1. Circle K has points V, C, and Z on the circle, and ΔVCZ is inscribed in circle K. $m\widehat{VC} = (9x + 23)°$, $m\angle CVZ = 36°$, and $m\angle VZC = (7x - 21)°$.



   a. What is the value of x?

   b. What is the $m\angle VCZ$?

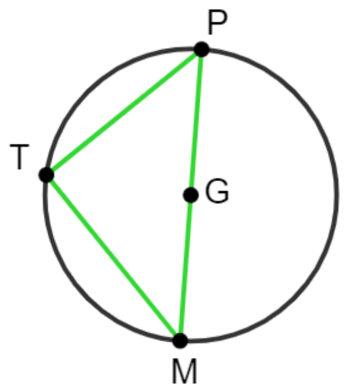
2. Circle A has points N, D, B, and S on the circle, and ΔSDN is inscribed in Circle A. $m\widehat{ND} = (7x - 7)°$, $m\widehat{DBS} = (23y - 5)°$, $m\angle NSD = (5x - 23)°$, and $m\angle DNS = (8y + 29)°$.



Complete the table.

Angle Measures	Arc Measures
$m\angle NSD =$	$m\widehat{ND} =$
$m\angle SND =$	$m\widehat{DBS} =$
$m\angle NDS =$	$m\widehat{NS} =$

3. Circle G has diameter $\overline{MP}$ and ΔMTP inscribed in the circle. $m\angle TMP = (6x - 8)^{\circ}$ and $m\angle TPM = (4x + 13)^{\circ}$.



a. Find the value of x.

b. What is the $m\angle TMP$?

c. What is the $m\angle TPM$?

11.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned about solving problems involving triangles inscribed in circles.